12 CONSTRUCTION OF BASIC LOGIC GATES

Aim:

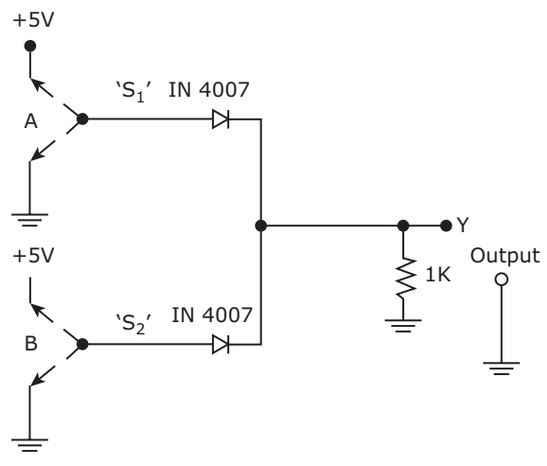
To Construct Logic Gates AND, OR, NOT by using Diodes, Transistor and Verify their Truth Table.

Apparatus / Components required:

S.No	Name of the apparatus / components	**Range / Value**	**Quantity**
1.	DC Regulated power supply	0-30V 1 A	3 No
2.	Diodes	IN 4007	4 No
3.	Transistor	BC 548	1 No
4.	LED	-	3 No
5.	Resistor	1K Ω,10K Ω, 200Ω	3 Nos
6.	Switches	SPDT	2 No
7.	Bread board	-	3 Nos
8.	Wire	Single strand	Required quantity

Circuit diagram

OR Gate



TRUTH TABLE

OR - Gate		
Input		Output
A	**B**	**Y=A+B**
0	0	0
0	1	1
1	0	1
1	1	1

Fig 12.1 **OR Gate**

Procedure:

1. Connect the components as per the OR, AND, NOT gate circuit in the bread board without loose contact .
2. Power supply voltage should not exceed 5V.
3. Verify the relevant truth table.

EE_Practicals_English Version.indd 264 11/10/2020 12:17:48 PM